

Why does sASM work — and only when the subdomain solve is inexact?

An interpretation through information propagation and overlap over-counting

Minimal 1D / 2D Laplace experiment (all numerics in PETSc): BASIC vs. sASM additive Schwarz, with *exact* (Cholesky) and *inexact* (ICC(0)) subdomain solves.

1D: 256 DOF, 8 subdomains. 2D: 64^2 grid, 4×4 subdomains.

The result that frames everything

CG iterations to $\|r\|/\|b\| < 10^{-8}$, same problem, only the subdomain solver changes

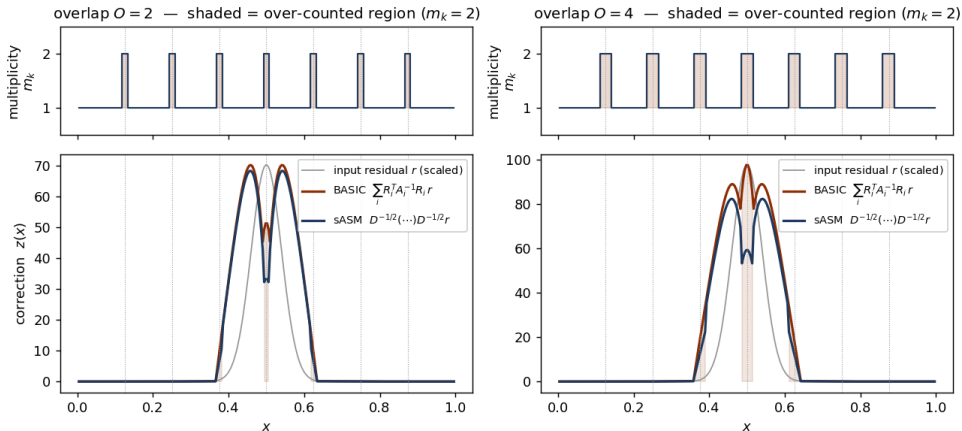
	1D Laplace		2D Laplace	
	exact	ICC(0)	exact	ICC(0)
BASIC	15	15	23	102
sASM	25	25	27	67

- ▶ **sASM beats BASIC only in one cell: 2D with an inexact solve (67 vs 102).**
Everywhere else sASM $\approx$ or slightly worse than BASIC.
- ▶ 1D “ICC” = exact: a tridiagonal matrix has no fill, so ICC(0) is the complete Cholesky.
Hence 1D cannot show the anomaly at all.
- ▶ **Question:** what makes the *inexact* solve bad, and why does sASM fix it precisely there?

Mechanism, part 1: BASIC over-counts the overlap

one preconditioner apply $\sum_i R_i^\top A_i^{-1} R_i r$ to a localized residual

Fig. 1 — One preconditioner apply: BASIC over-counts the overlap; sASM normalizes it

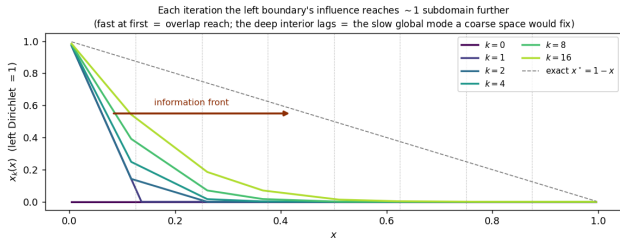


► In the overlap a DOF belongs to m_k subdomains, so its correction is *added* m_k times.

► BASIC (1.1) does not normalize the overlap, while sASM (1.1) does, by multiplying by $1/\sqrt{m_k}$.

Mechanism, part 2: information propagates one subdomain per iteration

g. 2 — Information propagation in one-level Schwarz: a bounded distance per iteration $\Rightarrow O(\#\text{subdomains})$ ite



- ▶ Laplace is globally coupled, but each Schwarz step solves only *local* problems.
- ▶ So the boundary's influence advances ~ 1 subdomain per iteration (fast at first = overlap reach).
- ▶ The deep interior lags — that slow global mode is what a coarse space (two-level) would remove.
- ▶ **One-level cost is $O(\#\text{subdomains})$ iterations**, independent of BASIC vs sASM. Overlap helps by speeding this propagation.

Q1 — Why is the inexact (ICC) solve bad?

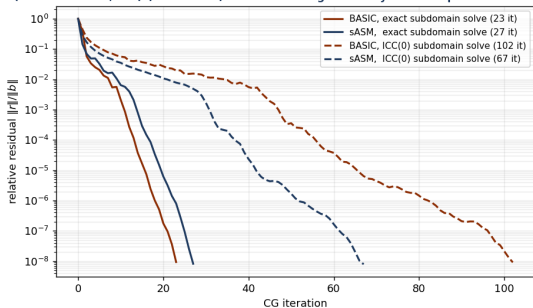
the abstract Schwarz bound: the two factors *multiply*

$$\kappa(\mathbf{M}^{-1}\mathbf{A}) \leq C_0^2 \cdot \underbrace{\omega}_{\substack{\text{(A) inexactness} \\ \text{exact}=1, \text{ ICC}>1}} \cdot \underbrace{N_c}_{\substack{\text{(B) over-count} \\ \text{(colouring / } m_k\text{)}}}$$

- ▶ **Exact** ($\omega=1$): each local correction is the *exact* local solution; over-counting only lifts $\lambda_{\max} \leq N_c$ — bounded, CG tolerates it. *No anomaly* (table: 23 vs 27).
- ▶ **ICC** ($\omega>1$): each local correction carries an *error*, and BASIC **adds** m_k **of these errors** in the overlap. They do not cancel — they accumulate:
 - ▶ $\lambda_{\max} \leq \omega N_c$: the inexactness is *multiplied* by the over-count;
 - ▶ wider overlap makes the block bigger, so fixed-fill ICC is a worse approximation ($\omega \uparrow$) *and* $N_c \uparrow$ — so κ *rises* with overlap (the anomaly).
- ▶ The accumulated, amplified error **piles up at the subdomain seams** and the outer CG must clear it.

Q1 — the evidence

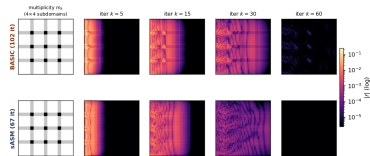
Fig. 3 — 2D Laplace: sASM beats BASIC ONLY with an inexact solve (exact: 23 vs 27; ICC(0): 67 vs 102) — over-counting bites only when coupled with inexact



Residual history. Exact: BASIC $\approx$ sASM (both fast). ICC: BASIC 102 blows up.

“Inexact is bad” = the local error is **counted** m_k **times in the overlap** ($A \times B$), accumulating into seam residual that costs iterations — and worsens as overlap grows.

Fig. 4 — ICC(0) subdomain solve: residual $|r|$ at MATCHED CG iterations. BASIC's residual lingers at the overlap seams; sASM clears it $\rightarrow$ 67 vs 102 iters

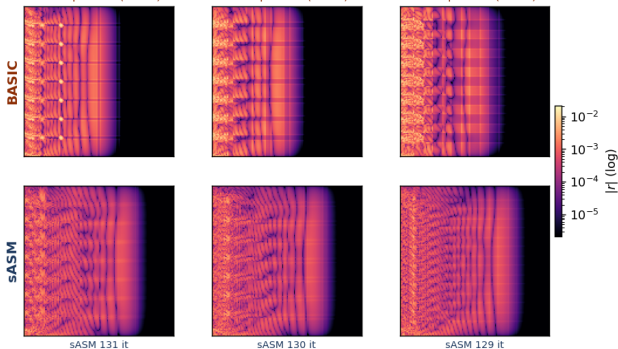


ICC residual field at matched CG iterations: at $k=60$ BASIC is still bright (error not cleared), the seams lit up by the over-counted ICC error.

Q1, deeper — the seam error grows with overlap

2D Laplace, 128^2 , 8×8 subdomains, ICC(0); residual $|r|$ at fixed iteration $k=40$

Fig. 5 — Overlap sweep (128^2 , 8×8 , ICC(0), residual at fixed $k=40$): BASIC's residual sits in the over-counted seams; the over-counted region (and the residual share trapped there, $0.45 \rightarrow 0.51 \rightarrow 0.64$) widens with O ; sASM clears it at every O
overlap $O=1$ (182 it) overlap $O=2$ (195 it) overlap $O=4$ (185 it)

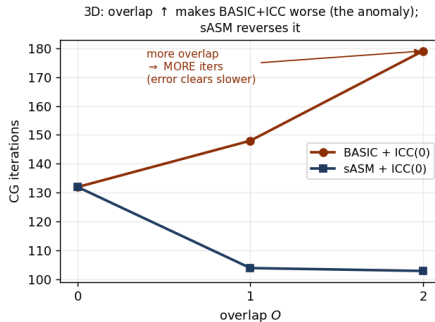
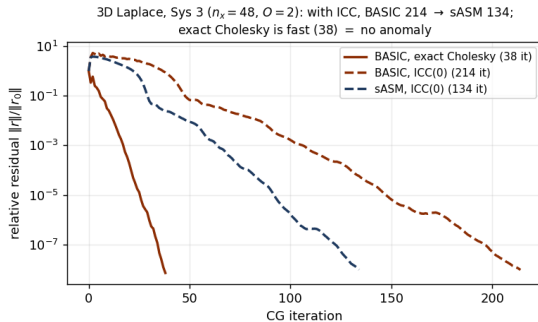


- ▶ The over-counted seam region *widens* with overlap, and the residual share trapped there grows $0.45 \rightarrow 0.51 \rightarrow 0.64$.
- ▶ sASM is better at every overlap (iters BASIC 182/195/185 vs sASM 131/130/129).
- ▶ Honest note: in 2D the iteration count vs O is non-monotone (ICC is only mildly inexact in 2D). The clean monotone “overlap $\uparrow \Rightarrow$ iter $\uparrow$ ” is a 3D effect $\rightarrow$ next slide.

The same effect on the real 3D problem (Sys 3)

mixed Dirichlet/Neumann Laplace, $n_x=48$, 4 ranks — the actual torso-type solve

Fig. 6 — The same effect on the real 3D Sys 3 (mixed Dirichlet/Neumann Laplace)



- ▶ **Left:** ICC — BASIC 214 slow, sASM 134 recovers; exact Cholesky 38 (no anomaly, the reference).
- ▶ **Right:** the clean monotone anomaly — BASIC+ICC 132 $\rightarrow$ 148 $\rightarrow$ 179 (more overlap, more iters = error clears slower); sASM 132 $\rightarrow$ 104 $\rightarrow$ 103 reverses it.

Q2 — Why does sASM overcome it?

$$\mathbf{M}_{\text{sASM}}^{-1} = \mathbf{D}^{-1/2} \left(\sum_i \mathbf{R}_i^\top \mathbf{A}_i^{-1} \mathbf{R}_i \right) \mathbf{D}^{-1/2}, \text{ partition of unity } \sum_i d_i^2 = 1$$

$$\text{BASIC: } \lambda_{\max} \leq \omega N_c \quad \xrightarrow{\mathbf{D}^{-1/2}(\cdot)\mathbf{D}^{-1/2}} \quad \text{sASM: } \lambda_{\max} \leq \omega$$

- ▶ The $\mathbf{D}^{-1/2}$ sandwich normalizes the overlap so each DOF's correction is counted **once**: the over-count factor N_c is **removed** (Fig. 1, blue is flat).
- ▶ **This breaks the multiplication.** The inexactness ω is no longer multiplied by N_c : the ICC error in the overlap is counted once, *not accumulated* $\Rightarrow$ no seam pile-up.
- ▶ Result (ICC): the seam residual clears (Fig. 4, sASM nearly dark at $k=60$); CG recovers **102** $\rightarrow$ **67**.
- ▶ **sASM does not fix the inexactness** (ω is still > 1) — it stops the over-count from *amplifying* it. So $\kappa \sim C_0^2 \omega$, and overlap helps again.

What does “exact / inexact subdomain solve” actually solve?

$A_i = R_i A R_i^\top$ — the global Laplace restricted to subdomain i (with overlap)

- ▶ A_i implicitly imposes **homogeneous Dirichlet** on the *artificial cut* between subdomains (the cross-interface coupling is dropped), and keeps the physical BC on faces that touch the real boundary.
- ▶ **Exact solve** A_i^{-1} (Cholesky) = *exactly* solving a local Dirichlet–Laplace subproblem.
 $\omega = 1$.
- ▶ **Inexact solve** ICC(0) = exactly solving a *perturbed* operator $\tilde{A}_i = A_i - E_i$, where E_i is the fill that ICC(0) *drops*. Equivalently, the same Dirichlet subproblem solved only approximately. This E_i **is** the source of $\omega > 1$, and its error sits in the overlap/seam.

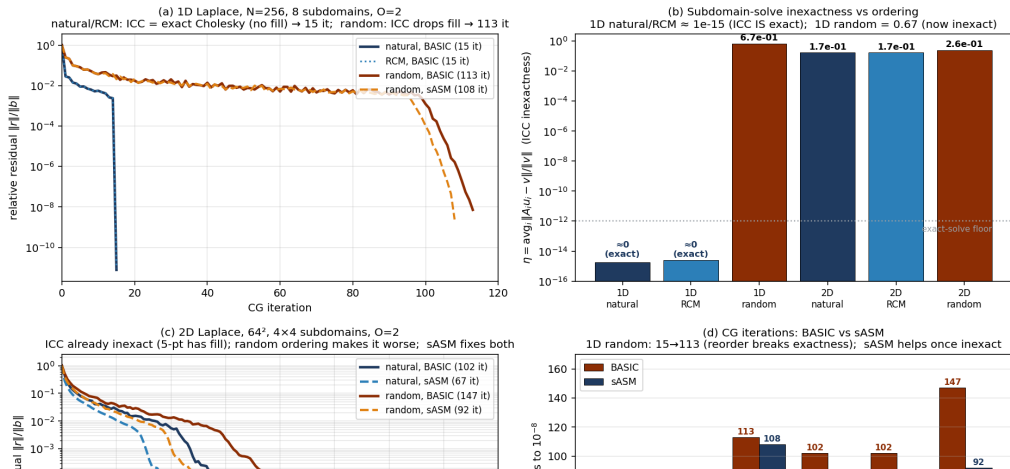
So “inexact” is not mysterious

ICC(0) keeps only A_i ’s nonzero pattern and throws away all fill-in. Whether that loses anything depends entirely on the **ordering** of the unknowns — next slide.

Reordering the unknowns turns the ICC inexact

same subdomains / overlap / weights — only the ICC *ordering* changes (natural / RCM / random)

Fig. 7 — Reordering the subdomain unknowns turns the ICC factorization inexact (Q1: 1D natural ICC = exact; random ICC $\neq$ exact)



Conclusion

The one-sentence picture

BASIC counts the overlap twice (Fig. 1). With *exact* solves that is harmless — it only lifts $\lambda_{\max}$, which CG tolerates (Fig. 3, solid). With an *inexact* ICC solve the local error is counted m_k times, so “inexactness $\times$ over-count” accumulates at the seams (Fig. 4) and convergence degrades (Fig. 3, dashed, 102). sASM’s $\mathbf{D}^{-1/2}$ partition of unity removes the over-count, breaking that product, so the inexact corrections are no longer amplified and CG recovers (102 $\rightarrow$ 67).

- ▶ **Q1:** inexactness is bad because BASIC *multiplies* it by the overlap over-count N_c ; and both grow with overlap, so iterations rise with overlap.
- ▶ **Q2:** sASM removes N_c (partition of unity), so the inexactness is no longer amplified — it helps *exactly* when the solve is inexact, and is neutral when it is exact.
- ▶ Information always propagates ~ 1 subdomain/iteration (Fig. 2): a one-level cost, untouched by either method.